

# Einwirkung

## Temperatur

```
from engicalc import *  
import numpy as np
```

Parameter

```
h = 0.95*m  
alpha_T = 10*10**-6 / K  
Delta_T1k = 20 *K  
Delta_T21k = 12 *K  
Delta_T22k = -4 *K  
  
Cell(show_expression=False, rows = 1)
```

$$h = 0.95 \text{ m}$$

$$\alpha_T = \frac{0.0}{\text{K}}$$

$$\Delta_{T1k} = 20 \text{ K}$$

$$\Delta_{T21k} = 12 \text{ K}$$

$$\Delta_{T22k} = -4 \text{ K}$$

Resultierende Dehnungen und Krümmungen

```
eps_x = (Delta_T1k * alpha_T).to(per mille)  
chi_T21 = (-Delta_T21k * alpha_T / h).to(1/km)  
chi_T22 = (-Delta_T22k * alpha_T / h).to(1/km)  
Cell(show_expression=False, rows = 1)
```

$$\varepsilon_x = 0.2 \text{ ‰}$$

$$\chi_{T21} = -\frac{0.13}{\text{km}}$$

$$\chi_{T22} = \frac{0.04}{\text{km}}$$